SKY TECH ACADEMY
ROPE ACCESS TRAINING CENTER

OILFIELD | INDUSTRY | SAFETY | EXPERTISE

INDUSTRIAL ROPE ACCESS TRAINING PROGRAM

Days 1 through 4 | SPRAT & IRATA Certification

DAY 1	DAY 2	DAY 3	DAY 4
Foundations	Regulations	Anchor Systems	Practice & Review

Houston, Texas | Serving Petrochemical | Oil & Gas | Construction | Wind Energy

TRAINING DAY 1 Industrial Rope Access Training

Classroom Rules & Expectations

- Safety First — safety is the top priority at all times
- Report all injuries to the instructor immediately, no matter how minor
- Limit cell phone use — keep phones silent during training sessions
- Store personal items and lunch in designated areas; do not consume others' food
- Restrooms are available, please respect designated facilities
- Remove all rope access equipment before entering office areas
- Clean up after yourself and maintain a tidy training environment
- Maintain a positive attitude, participate actively, and have fun

Introduction — Instructor & Student Introductions

Instructor

- Introduction of instructor
- Rope access experience:
- Training and instructional experience:

Students — Please Introduce Yourself

- Your name and company
- Your role at work
- Which training scheme and certification level you are attending
- Your previous rope access experience (if any)

Day 1 Agenda

Session	Topics Covered
Course Overview & Expectations	Introduction to safety principles and hazard awareness; overview of SPRAT & IRATA training schemes
Equipment & Safety	Equipment familiarization; inspection procedures (pre-use and formal checks); kitting up; function check
Roles & Worksite Awareness	Roles and responsibilities; housekeeping and site awareness; evacuation routes; introduction to JSA
Administration	Required paperwork and documentation; training and certification requirements
End of Day	Post-training discussion and review; equipment clean-up and storage; final housekeeping

Key Takeaway: *Safety, preparation, and correct procedures are essential at every stage. What you learn here forms the foundation for safe rope access work.*

Day 1 Learning Outcomes

By the end of Day 1, participants will be able to:

- Understand the principles of industrial rope access systems
- Identify key components of Personal Protective Equipment (PPE)
- Recognize applicable industry standards (SPRAT & IRATA)
- Demonstrate safe equipment inspection practices
- Understand roles, responsibilities, and safety expectations

Introduction to Rope Access

Industrial rope access is a safe and efficient method for working at height, allowing technicians to reach areas that are otherwise difficult or inaccessible. Its effectiveness comes from the strict application of proven safety principles.

Core Safety Principles

1. Redundant Systems

- Use of a primary working rope alongside an independent backup (safety) rope
- Provides fail-safe fall protection in case one system is compromised

2. Continuous Attachment — 100% Tie-Off

- Technicians are secured at all times while working at height
- Eliminates exposure to unprotected falls

3. Controlled Access and Descent

- Enables precise positioning and movement during work
- Maintains full control during ascent, descent, and task execution

> **Key Takeaway:** *Rope access is one of the safest height-access methods because it is built on redundancy, constant connection, and controlled movement.*

Training & Documentation Checklist

All participants must complete the following documentation prior to attending training to ensure compliance, safety, and eligibility.

1. Mandatory Agreements — Liability Agreement

- Complete the Release of Liability and Assumption of Risk Agreement
- Print your name where indicated
- Initial each clause to confirm understanding and acceptance
- Sign and date the document

2. Required Supporting Documents

- Copy of government-issued identification (ID)
- SPRAT Evaluation Session Candidate Affidavit
- SPRAT Privacy Acknowledgement
- IRATA Candidate Disclaimer and Liability Release
- Certification of Physical Condition

3. Important Requirements

- All documentation must be accurately completed and signed where required
- Must be submitted before participation in any practical training

Incomplete or missing documentation may result in delayed or denied participation.

Training & Certification — SPRAT vs. IRATA Comparison

Topic	SPRAT	IRATA
Operations	Safe Practices for Rope Access Work	International Code of Practice (ICOP)
Certification	Certification Requirements for Rope Access Work	Training, Assessment & Certification Scheme (TACS)
Assessment Day	Evaluation Requirements	Training, Assessment & Certification Scheme
Training Required	No — Recommended (Affidavit required)	Yes — 32 hours minimum
Upgrade Requirement	500 hours + 6 months (L1→L2)	1,000 hours + 1 year (L1→L2)
Refresher	Recommended	8 hours if inactive for 6+ months
Supervision	Level 2 in limited circumstances	Level 3 only
Certification Cycle	3 years	3 years

Training Safety Rules & Requirements

Safety First — Always

- Treat the training environment as a real worksite at all times
- Maintain a safety-first mindset in every activity

Personal Protective Equipment (PPE)

- Helmets are mandatory inside designated exclusion zones
- Gloves are recommended and available for purchase: Leather gloves ~$25 | General work gloves ~$10

Facilities & Equipment

- First Aid Kit — located within the training center; inform instructor of any items used
- Drinking water — readily available; stay hydrated throughout training

Worksite Procedures

- A Job Safety Analysis (JSA) must be completed every morning
- All participants must sign the JSA before starting work
- Follow the JSA and work in accordance with identified hazards and control measures

> **Key Takeaway:** *Follow all rules, use equipment properly, stay alert and aware at all times. Safe behavior is mandatory — no exceptions.*

Practical Exercises — Day 1

Level 1	Level 2	Level 3
Kitting up	Horizontal Aid Climb	Level 2 exercises
Use of Equipment	Vertical Aid Climb	Pitch head / Platform haul
Changeover	Rescue kits	Anchors pre-rigged to lower
Pass Deviation (S–Single, I–Single/Double)	Lowering Systems	Rescue kits
Transfer	Pitch head / platform haul	Hauling and Lowering through knots
Knots (Figure 8, Figure 9)		

Time to Climb!

TRAINING DAY 2 Working at Height — Regulations & Fall Protection

Working at Height — Global Regulatory Overview

Working at height is governed by different regulatory bodies worldwide. While regulations differ by region, core safety principles remain similar.

Region	Governing Body / Standard
United States	OSHA — supported by ANSI Z359 standards
United Kingdom	HSE — British Standards (BS)
European Union	EN Standards
Canada	CCOHS — CSA Standards

OSHA & Workplace Safety Overview

What is OSHA?

- OSHA = Occupational Safety and Health Administration
- U.S. authority responsible for developing and enforcing workplace safety laws
- Applies to all workplaces in the United States

OSHA Standards Structure

- 29 CFR 1910 — General Industry
- 29 CFR 1926 — Construction

These are the main regulatory frameworks for workplace safety in the United States.

OSHA vs. ANSI — How They Work Together

OSHA	ANSI
Legal authority in the U.S.	Develops consensus safety standards
Compliance is mandatory	Voluntary by definition
Defines required workplace safety rules	Provides detailed technical guidance
Establishes minimum requirements	Defines the 'how-to' for compliance

Key Takeaway: *OSHA = What you must follow (law). ANSI = How you safely meet those requirements. Real compliance requires both working together.*

ANSI Z359 Fall Protection Standards

Standard	Title	Relevance
Z359.1	Fall Protection Code	Minimum guidelines for fall protection programs; systems and components
Z359.2	Managed Fall Protection Program	Employer requirements; policies, roles, training, program management
Z359.3	Rescue Systems	Assisted rescue and self-rescue system requirements
Z359.11	Full Body Harnesses	Design, performance, and strength requirements for harnesses
Z359.14	Self-Retracting Lifelines	SRL performance and safety requirements

Key Definitions — Fall Protection

Fall Protection

A backup system designed for workers at height when there is a risk of losing balance or falling. Purpose: to control a fall if it occurs, eliminate or reduce injury risk, and ensure worker safety.

1. Fall Arrest

Designed to safely stop a person already in free fall. Requires a full body harness, energy-absorbing system, and secure anchor point. Purpose: minimize injury by controlling and stopping the fall.

2. Work Positioning

Supports a worker on a vertical surface (wall, tower, structure) for hands-free work. Uses harness, ropes, lanyards, and connectors. Purpose: provide stable positioning to perform work safely.

3. Rope Access (Relation to Work Positioning)

Rope access is a form of work positioning system. Technicians are supported by ropes, maintain hands-free capability, and use controlled movement for access and positioning.

4. Fall Restraint

Prevents the worker from reaching a fall hazard by restricting movement. Purpose: eliminate the possibility of a fall before it can occur.

5. Managed Fall Arrest Program

A formal employer program to protect employees exposed to fall hazards. Includes procedures, equipment selection, training, and inspection processes.

> **Key Takeaway:** *Fall Arrest = Protection after a fall starts. Work Positioning = Prevention and control while working. Rope Access = Advanced work positioning with controlled movement. Fall Restraint = Prevents access to hazard.*

Fall Arrest — ANSI Z359.2

Fall arrest systems stop a worker after a fall has occurred, preventing ground contact. The harness and connecting system absorb energy and safely stop the fall.

Fall Arrest Attachment Points & Force Limits

Connection Point	Peak Arrest Force	Average Force	Max Free-Fall Distance
Sternal (Front)	900 lbs (4 kN)	450 lbs (2 kN)	2 feet
Dorsal (Back)	1,800 lbs (8 kN)	900 lbs (4 kN)	6 feet (unless otherwise certified)

Key Takeaway: *Sternal = lower forces, shorter falls. Dorsal = higher limits, longer allowable fall distance. Selection depends on application, system design, and safety requirements.*

Work Positioning — ANSI Z359.2

Work positioning systems hold a worker in place while preventing access to fall hazards. The worker is restrained from reaching an edge — used for support and stability, not fall arrest.

Equipment Requirements

- D-Rings: Minimum Breaking Strength (MBS) = 5,000 lbs (22.2 kN)
- Buckles: Minimum Breaking Strength = 4,000 lbs (17.8 kN)
- Lateral side D-rings must ALWAYS be used in pairs — never individually
- A secondary fall protection system is required when exposure to a fall hazard exists

Work Positioning Anchorage Requirements

- Certified anchorage: must support at least 2× the foreseeable force applied
- Non-certified anchorage: minimum 3,000 lbs (13.3 kN) per worker
- Only one worker may be attached per segment at a time
- Anchor points must be properly positioned to prevent reaching a fall hazard

Key Takeaway: *Work positioning = fall prevention, not fall arrest. Worker must be kept away from the hazard. A backup system is required if any fall risk exists.*

Fall Factor

Fall Factor indicates the severity of a potential fall. It is calculated by dividing the distance of fall by the length of rope in the system.

Fall Factor = Length of Fall ÷ Length of Rope

Fall Factor	Fall Distance	Rope Length	Severity
FF = 0	0 ft	3 ft	☑ Ideal — no fall
FF = 1	3 ft	3 ft	⚠ Moderate
FF = 2	6 ft	3 ft	▨ Maximum risk

Key Takeaway: *Keep the backup device as high as practical to minimize fall factor. The goal in rope access is always FF = 0 or as close as possible.*

Fall Protection Saves Lives

Falls from height are one of the leading causes of serious injury and fatalities in the workplace. Most incidents happen because of incorrect equipment use, poor anchor selection, or lack of inspection or training.

Fall protection is not optional. Small mistakes at height can have serious or fatal consequences.

Practical Exercises — Day 2

Level 1	Level 2	Level 3
Repeat Day 1 activities	Rigging Rope Systems	Use of Tensioned Lines
Pass short re-anchor (I)	Rope and Sling Protection	Complex Rescue Scenario
Pass long re-anchor (S)	Pre-rigged to Lower Systems	Work Scenario
Pass Knots	Retrievable Rigging	
Fixed aid climb	Load Sharing Anchors	
Mobile aid climb	Cross Haul	
Climbing with Fall Arrest Lanyards (I)	Tensioned Lines (I)	
Knots: Alpine, Stop, Join	Rigging Fall Restraint Lines (I)	
	Knots: Prusik, Munter	

Time to Climb!

TRAINING DAY 3 Anchor Systems, Load Angles & Load Limits

Anchor Systems — Critical Safety Principles

Fundamental Rule: No anchor = No safe system

Anchors are the foundation of all rope access work at height. Failure of an anchor results in total system failure.

Core Requirements

- Anchors must be strong enough for the intended load
- Anchors must be the correct type for the task
- Anchors must be connected to a structurally sound anchorage

System Rules — Non-Negotiable

- Always use TWO independent anchors
- Anchors must be separate and independent
- Failure of one anchor must NOT affect the second

Positioning Rules

- Anchors should be placed to minimize fall distance
- Anchors should reduce swing fall risk

Load Management

- Distribute loads effectively — use load sharing when applicable
- Avoid overloading a single anchor point

Protection & Setup Rules

Anchors must be protected from: sharp edges, heat, abrasion, and chemicals. They must be installed to prevent movement and avoid accidental disconnection.

> **Key Takeaway:** *Use two independent anchors. Select and position anchors carefully. Protect and manage loads properly. Safe rope access starts with proper anchor setup and control.*

Load Angles — Anchor Angle & Load Distribution

The included angle is the angle between two anchor lines (rope-to-rope), measured on the inside of the system. As the included angle increases, force on each anchor increases and load sharing becomes less efficient.

Zone	Angle	Force on Each Anchor	Safety Status
Preferred Zone	≤ 90°	50–71% of load	☑ Best load distribution
Critical Zone	≈ 120°	~100% of load	⚠ Each anchor at full load
Danger Zone	> 120°	200–300%+ of load	▨ High risk of anchor failure

Included Angle Force Reference Table

Angle	Force on Each Anchor Leg
0°	50% of total load
45°	57% of total load
90°	71% of total load
120°	100% of total load (critical)
150°	200% of total load (dangerous)
160°	300% of total load (extreme danger)

★ **Key Takeaway:** *Smaller angle = Safer system. Keep angles ≤ 90° whenever possible. Never approach or exceed 120°. Proper anchor angle control is essential for safe load distribution.*

WLL vs. MBS — Understanding Load Limits

MBS — Minimum Breaking Strength	WLL — Working Load Limit
Maximum load before failure (failure point)	Maximum safe load during normal operations
Determined under controlled test conditions	Based on a safety factor applied to MBS
Absolute limit — NOT for actual use	Includes safety margin to prevent failure
Safety limit reference only	Used in all real-world operations

General Safety Factors

- Solid metal components (carabiners, connectors, anchor hardware): WLL ≈ 20% (1/5) of MBS
- Fabric & mechanical components (ropes, slings, harnesses, descenders, ascenders): WLL ≈ 10% (1/10) of MBS

Equipment SWL Reference Table

Equipment	MBS	SWL	Notes
Anchor (non-engineered)	5,000 lbf	1,000 lbf	Per worker for fall arrest
Carabiner	27 kN (6,070 lbf)	1,213 lbf	20% safety factor
Rope (with knot)	6,000 lbf	420 lbf	10% with knot reduction
Rigging Plate Paw	36 kN (8,095 lbf)	1,618 lbf	Metal component
Pulley (Body)	23 kN (5,170 lbf)	1,034 lbf	Metal component
I'D S Descender	14 kN (3,148 lbf)	330 lbf	Mechanical/fabric factor

Conversion: 1 kN = 224.8 lbf | 1 kg = 2.2 lbs

Key Takeaway: *MBS = Breaking point. WLL = Safe working limit. Never load equipment near its MBS. Always operate within WLL and consider additional forces: shock load, angles, and dynamic movement.*

Practical Exercises — Day 3

Level 1	Level 2	Level 3
Repeat Day 2 activities	Rescue Through Knots (S)	Rescue Through Knots (I)
Negotiate Edge	Rescue from Horizontal Aid	Rescue Past Deviation (S–Single, I–Double)
Rope and Sling Protection	Rescue from Ascent (I)	Mid Transfer Rescue
Anchor Selection	Single Deviation Rescue (I)	Rescue Through Re-anchor
Simple Structural Anchor	Rope to Rope Transfer	Rescue from Short Link Aid
Load Sharing Anchors (< 1m)	Rescue from Fall Arrest (I)	
Hauling and Lowering		

Time to Climb!

TRAINING DAY 4 Practice, Catch-up & Evaluation Preparation

Practice & Performance Guidance

Day 4 is dedicated to consolidating all skills from Days 1–3, catching up on any techniques requiring additional practice, and preparing for the certification evaluation.

During Training

- Practice and refine all rope access techniques from previous days
- Manage your energy — do not overexert yourself
- Prepare for continued performance in your evaluation

Learning Actions

- Observe other participants — learn from different techniques and approaches
- Use downtime effectively: practice knots and rigging at ground level
- Ask the instructor for feedback on areas needing improvement

Skill Development Focus

- Focus on challenging techniques identified during Days 1–3
- Avoid over-focusing on one task — build overall competence across all skills
- Aim to perform each skill smoothly, efficiently, and safely

Personal Readiness

- Get adequate rest each night during the training program
- Rehydrate properly after training sessions
- Maintain physical readiness for each session and the final evaluation

Key Takeaway: *Improvement depends on practice, observation, and recovery. Consistent effort and proper recovery lead to better performance and safer work at height.*

Day 4 Review — Full Skills Checklist

Level 1 Skills	Level 2 Skills	Level 3 Skills
Kitting up correctly	Horizontal & Vertical Aid Climb	All Level 1 & 2 skills
Equipment inspection & function check	Rigging rope systems	Use of tensioned lines
Controlled descent (changeover)	Rope & sling protection	Complex rescue scenarios
Pass deviation (single)	Load sharing anchors	Pitch head / platform haul

Level 1 Skills	Level 2 Skills	Level 3 Skills
Transfer between ropes	Pre-rigged to lower systems	Hauling & lowering through knots
Knots: Fig. 8, Fig. 9	Retrievable rigging	Mid transfer rescue
Negotiate edge	Cross haul	Rescue through re-anchor
Simple structural anchor	Tensioned lines (I)	Rescue from short link aid
Load sharing anchors (<1m)	Rescue through knots (S)	Rescue through knots (I)
Hauling and lowering	Rope to rope transfer	Rescue past deviation
Knots: Alpine, Stop, Join	Knots: Prusik, Munter	Rescue from fall arrest

Evaluation Reminder — What to Expect

Your certification assessment evaluates practical skills used in real rope access work. Remember:

- Performance-based — there is flexibility in how tasks are completed as long as they are performed safely
- Each skill is graded: Pass (P) / Discrepancy (D) / Fail (F)
- One Fail = automatic overall failure
- Three Discrepancies = automatic overall failure
- Immediate failure: relying on single rope, improper backup device use, requiring rescue to complete exercise

Discrepancies are issued for minor errors such as an unlocked carabiner, unsecured helmet strap, or insufficient friction during a loaded descent. These are taken seriously as they reflect overall safety awareness.

Key Takeaway: *The techniques you've learned must not only be retained for the evaluation but applied consistently in the field. Certification is a demonstration of real competence, not just passing a test.*

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